

Minimal additive codes

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Abstract

Additive codes generalize linear codes by relaxing the field of linearity. In this paper, we initiate the study of minimal additive codes from a geometric perspective. We introduce the notion of additive strong blocking sets as a generalization of outer strong blocking sets, and establish a one-to-one correspondence between equivalence classes of minimal additive codes and equivalence classes of additive strong blocking sets. Moreover, we provide a first sufficient condition for their existence, laying the foundation for further investigation into their parameters, constructions, and applications.

Key Words: additive codes; minimal codes; strong blocking sets.

1 Introduction

A linear code $\mathcal{C}$ is a k -dimensional subspace of the n -dimensional vector space over the finite field $\mathbb{F}_q$, endowed with a distance, generally, the Hamming one. The minimum distance d of $\mathcal{C}$ is the least among the distances of every two vectors in the code. It is well-known that a linear code corresponds to a *projective system*, i.e. a multiset of n points in the projective space $\text{PG}(k-1, q)$, such that each hyperplane contains at most $n-d$ of them. Additive codes generalize the classical notion of linear codes by relaxing the requirement of linearity over the ambient alphabet. More specifically, an additive code of length n over $\mathbb{F}_{q^h}$ is defined as an $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^h}^n$. Such codes need not be linear over the extension field $\mathbb{F}_{q^h}$ itself, but they retain enough algebraic structure to allow geometric interpretations via subspaces in projective space. Recent works on additive codes introduce a geometric point of view by generalizing the concept of projective system. Indeed, additive codes can be seen as collections of n subspaces of $\text{PG}(k-1, q)$ of dimension at most $h-1$, such that each hyperplane contains at most

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$n - d$ such subspaces. Thanks to this approach, interesting results on the parameters of additive codes have been derived; see e.g. [Kur24, DK25, AB23, BGMZ25]. Several important families of codes arise naturally as additive codes. For example, there is some renewed interest in additive codes due to applications in the construction of quantum codes; see [DL24, DS25, Gra21].

Minimal linear codes have attracted the attention of many researchers in recent years not only for their application in secret sharing schemes ([Mas93]), but also for their combinatorial properties. In fact, in [ABN22] and in [TQLZ21], it was independently shown that minimal codes are in one-to-one correspondence with projective systems that form *strong blocking sets*. This correspondence has been intensively exploited for finding constructions and bounds; see for example [ABNR22, HN21, BB23, ABDN24].

In [ABN24], the notions of *outer strong blocking sets* and *outer minimal codes*, have been introduced. These are sets of points whose field reduction is a strong blocking set and codes whose concatenation with minimal codes is minimal. We generalize these notions by weakening the field of linearity. In particular, we introduce the more general concepts of *minimal additive codes* and *additive strong blocking sets*. We illustrate a correspondence between them and we provide a first sufficient condition for their existence.

The paper is organized as follows. In Section 2, we provide the necessary background to the rest of the paper. In Section 3, we introduce the main concepts of minimal additive codes and additive strong blocking sets. We initiate their study, providing a one-to-one correspondence between them. In Section 4, we provide a first existence result. We conclude in Section 5 with some future research directions.

2 Background

In this section, we revise some basic notions on additive codes, and we fix the notation for the rest of the paper. For more details we refer to [BGL23, Kur24, BP25, BGMZ25].

Notation 2.1. Let q be a prime power, $\mathbb{F}_q$ be the finite field of order q and $\mathbb{F}_{q^h}$ be the finite field extension of $\mathbb{F}_q$ of order h . The space $\mathbb{F}_{q^h}^n$ is the n -dimensional vector space over $\mathbb{F}_{q^h}$. All vectors are denoted as row vectors, unless otherwise stated. Let $\mathcal{B} = \{\alpha_1, \dots, \alpha_h\}$ be a basis of $\mathbb{F}_{q^h}$ over $\mathbb{F}_q$ and denote $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_h)^\top$. We denote by $\mathcal{M}_{r,s}(\mathbb{F}_q)$ the space of $r \times s$ matrices with entries in $\mathbb{F}_q$. For every $0 \leq \ell \leq n$, the number of ℓ -dimensional subspaces of $\mathbb{F}_q^n$ is denoted by the Gaussian binomial coefficient

$$\binom{n}{\ell}_q = \frac{(q^n - 1)(q^{n-1} - 1) \dots (q^{n-\ell+1} - 1)}{(q - 1)(q^2 - 1) \dots (q^\ell - 1)}.$$

Finally, we denote by $\text{PG}(k - 1, q)$ the projective space with underlying vector space $\mathbb{F}_q^k$.

An $[n, k/h]_q^h$ **additive code** $\mathcal{C}$ is a k -dimensional $\mathbb{F}_q$ -linear subspace of $\mathbb{F}_{q^h}^n$. This standard notation highlights the formal ratio k/h , which may or may not be an integer, and it is equal to $\log_{q^h} |\mathcal{C}|$. Given a vector $v \in \mathbb{F}_{q^h}^n$, we say that the **support** of v is the set $\text{supp}(v) := \{i \in \{1, \dots, n\} : v_i \neq 0\}$ and the **Hamming weight** of v is $\text{wt}(v) = |\text{supp}(v)|$. The Hamming weight naturally induces a metric on the space $\mathbb{F}_{q^h}^n$, known as **Hamming distance**, which is defined as $d(u, v) = \text{wt}(u - v)$, for every $u, v \in \mathbb{F}_{q^h}^n$. The **minimum Hamming distance** of an additive code $\mathcal{C}$ is defined, as usual, as

$$d(\mathcal{C}) := \min\{d(u, v) : u, v \in \mathcal{C}, u \neq v\}.$$

When the minimum distance d of $\mathcal{C}$ is known, we say that $\mathcal{C}$ is an $[n, k/h, d]_q^h$ additive code. A matrix $G \in \mathcal{M}_{k,n}(\mathbb{F}_{q^h})$ is a **generator matrix** for $\mathcal{C} \subseteq \mathbb{F}_{q^h}^n$ if it has full rank and its rows form an $\mathbb{F}_q$ -basis of $\mathcal{C}$. We say that $\mathcal{C}$ is **degenerate** if there exists a coordinate $i \in \{1, \dots, n\}$, such that $c_i = 0$ for every $c \in \mathcal{C}$. If $\mathcal{C}$ is not degenerate, we say that it is **nondegenerate**.

Definition 2.2. Two $[n, k/h, d]_q^h$ additive codes $\mathcal{C}, \mathcal{D}$ are **equivalent** if one can be obtained from the other by permuting the coordinate positions, or by applying an $\mathbb{F}_q$ -linear automorphism of $\mathbb{F}_{q^h}$ (regarded as $\mathbb{F}_q$ -vector space) in each coordinate position.

Recently, additive codes have been studied in connection to finite geometry. We recall here this point of view. For more details, we refer the interested reader to [BGL23, Kur24, BGMZ25, AB23].

Let $\mathcal{C}$ be a $[n, k/h, d]_q^h$ code and let $G \in \mathcal{M}_{k,n}(\mathbb{F}_{q^h})$ be a generator matrix for $\mathcal{C}$. We write

$$G = (\textcolor{red}{g}_1 \mid \textcolor{green}{g}_2 \mid \cdots \mid \textcolor{blue}{g}_n),$$

where $g_i = (g_{i,1}, \dots, g_{i,k})^\top \in \mathbb{F}_{q^h}^k$ is the i -th column of G . Each g_i is a vector with entries in $\mathbb{F}_{q^h}$, hence we can expand it over $\mathbb{F}_q$, with respect to the basis $\mathcal{B}$, i.e. $g_{i,j} = \sum_{t=1}^h b_{i,j}^t \alpha_t$, where $b_{i,j}^t \in \mathbb{F}_q$ for every i, j, t . This yields a matrix $\tilde{G} \in \mathcal{M}_{k,nh}(\mathbb{F}_q)$ with entries in $\mathbb{F}_q$, whose columns are grouped into n blocks of size h :

$$\tilde{G} = \begin{pmatrix} \textcolor{red}{b}_{1,1}^1 & \cdots & \textcolor{red}{b}_{1,1}^h & \textcolor{green}{b}_{1,2}^1 & \cdots & \textcolor{green}{b}_{1,2}^h & \cdots & \textcolor{blue}{b}_{1,n}^1 & \cdots & \textcolor{blue}{b}_{1,n}^h \\ \textcolor{red}{b}_{2,1}^1 & \cdots & \textcolor{red}{b}_{2,1}^h & \textcolor{green}{b}_{2,2}^1 & \cdots & \textcolor{green}{b}_{2,2}^h & \cdots & \textcolor{blue}{b}_{2,n}^1 & \cdots & \textcolor{blue}{b}_{2,n}^h \\ \vdots & & \vdots & \vdots & & \vdots & & \vdots & & \vdots \\ \textcolor{red}{b}_{k,1}^1 & \cdots & \textcolor{red}{b}_{k,1}^h & \textcolor{green}{b}_{k,2}^1 & \cdots & \textcolor{green}{b}_{k,2}^h & \cdots & \textcolor{blue}{b}_{k,n}^1 & \cdots & \textcolor{blue}{b}_{k,n}^h \end{pmatrix} \quad (1)$$

If $\mathcal{C}$ is nondegenerate, the columns of each block of $\tilde{G}$ span a non-zero $\mathbb{F}_q$ -subspace of $\mathbb{F}_q^k$ of dimension at most h . We can regard each of these spaces as a projective subspace of $\text{PG}(k-1, q)$ of dimension at most $h-1$. Let $\chi_G(\mathcal{C}) = \{\pi_1, \dots, \pi_n\} \subseteq \text{PG}(k-1, q)$ be the multiset of such projective subspaces. In

[BGL23, BP25], the link between $\mathcal{C}$ and $\chi_G(\mathcal{C})$ has been investigated, and, in particular, it is shown that the choice of G does not change the multiset $\chi_G(\mathcal{C})$, so we can simply use the notation $\chi(\mathcal{C})$, without specifying G . We recall the following definition from [BLP25].

Definition 2.3. Let $\chi = \{\pi_1, \dots, \pi_n\}$ be a multiset of subspaces of $\text{PG}(k-1, q)$. We say that χ is a h - $[n, k, d]_q$ **projective system** if $\dim(\pi_i) \leq h-1$ for every $i \in \{1, \dots, n\}$ and the spaces π_i are not contained in the same hyperplane. Moreover, we have that

$$n - d = \max_{\substack{H \subseteq \text{PG}(k-1, q), \\ H \text{ hyperplane}}} |\{i \in \{1, \dots, n\} : \pi_i \subseteq H\}|,$$

i.e. every hyperplane contains at most $n - d$ elements of χ and there exists an hyperplane that contains exactly $n - d$ elements of χ .

Definition 2.4. We say that two h - $[n, k, d]_q$ projective systems χ and ψ are **equivalent** if there exists a $\varphi \in \text{PGL}_k(\mathbb{F}_q)$, such that $\varphi(\chi) = \psi$.

The following is the generalization to the additive case of the well-known correspondence between linear codes and projective systems; see [TVN07].

Theorem 2.5 ([BLP25]). There is a one-to-one correspondence between equivalence classes of nondegenerate $[n, k/h, d]_q^h$ codes and equivalence classes of h - $[n, k, d]_q$ projective systems.

We briefly explain the correspondence. Let $\mathcal{C}$ be an $[n, k/h, d]_q^h$ additive code and $\chi(\mathcal{C})$ be defined as above. Then $\chi(\mathcal{C})$ is an h - $[n, k, d]_q$ projective system. Indeed, if $c \in \mathcal{C}$ and $a \in \mathbb{F}_q^k$ such that $c = aG$, the weight of c can be obtained as

$$\text{wt}(c) = \text{wt}(aG) = n - |\{i \in \{1, \dots, n\} : \pi_i \subseteq \langle a \rangle^\perp\}|.$$

Therefore, the minimum distance of the code $\mathcal{C}$ is

$$d = n - \max_{\substack{H \subseteq \text{PG}(k-1, q), \\ H \text{ hyperplane}}} |\{i \in \{1, \dots, n\} : \pi_i \subseteq H\}|.$$

In particular, every nonzero codeword $c = aG$ corresponds to the hyperplane $\langle a \rangle \subseteq \text{PG}(k-1, q)$.

Vice versa, let $\{\pi_1, \dots, \pi_n\}$ be an h - $[n, k, d]_q$ projective system. For each π_i , choose a $k \times h$ matrix G_i , such that $\text{colsp}_{\mathbb{F}_q}(G_i) = \pi_i$. Construct $G = (G_1 \alpha \mid \dots \mid G_n \alpha) \in \mathcal{M}_{k, n}(\mathbb{F}_{q^h})$, where α is as in Notation 2.1. Then $\mathcal{C} = \text{rowsp}_{\mathbb{F}_q}(G)$ is an $[n, k/h, d]_q^h$ additive code. The correspondence described above depends on the choices of generator matrices for $\mathcal{C}$ or bases for the π_i 's. However, it is well-defined at the level of equivalence classes: different choices yield equivalent codes and projective systems.

3 Minimal Additive Codes

In this section, we introduce the notion of *minimal additive codes* and we describe their geometric counterpart by generalizing the concept of outer strong blocking sets and outer minimal codewords. Outer minimal codes are $\mathbb{F}_{q^h}$ -linear subspaces of $\mathbb{F}_{q^h}^n$ whose concatenation with minimal codes is minimal. In [ABN24], it is shown that they are in one-to-one correspondence with outer strong blocking sets, which are sets of points in $\text{PG}(k/h - 1, q^h)$ whose field reduction is a strong blocking set.

An $\mathbb{F}_q$ -linear subspace $\mathcal{C}$ of $\mathbb{F}_q^n$ is said to be minimal if for every $c, c' \in \mathcal{C} \setminus \{0\}$, we have $\text{supp}(c') \subseteq \text{supp}(c) \Leftrightarrow c = \lambda c'$, for some $\lambda \in \mathbb{F}_q^*$. A strong blocking set in $\text{PG}(k - 1, q)$ is a multiset of points $\mathcal{M}$, not all lying on the same hyperplane, such that for all hyperplanes $H, H' \subseteq \text{PG}(k - 1, q)$, $\mathcal{M} \cap H \subseteq \mathcal{M} \cap H' \Leftrightarrow H = H'$. In [ABN22, TQLZ21] it is shown that (monomial) equivalence classes of minimal linear codes are in one-to-one correspondence with projectively equivalent classes of strong blocking sets. The correspondence is given as follows. Given a nondegenerate linear code $\mathcal{C}$ generated by $G \in \mathcal{M}_{k,n}(\mathbb{F}_q)$, define the set of projective points $\mathcal{P}_G(\mathcal{C}) \subseteq \text{PG}(k - 1, q)$ represented by the columns of G . Vice versa, given a set of points $\mathcal{S} \subseteq \text{PG}(k - 1, q)$, we construct the generator matrix $G_{\mathcal{S}}$ of a linear code with columns given by the coordinates of the points in $\mathcal{S}$. Since this correspondence does not depend on the choice of generator matrix, we simply write $\mathcal{P}(\mathcal{C})$, for denoting a projective system associated to $\mathcal{C}$.

Definition 3.1. Let $\mathcal{C}$ be an additive code. We say that a codeword $c \in \mathcal{C} \setminus \{0\}$ is **minimal** if $\forall c' \in \mathcal{C} \setminus \{0\}$ such that $\text{supp}(c') \subseteq \text{supp}(c)$ **and** $\forall i \in \text{supp}(c'), \frac{c'_i}{c_i} \in \mathbb{F}_q^*$ then $\exists \lambda \in \mathbb{F}_q^*$ such that $c' = \lambda c$.

Remark 3.2. Note that Definition 3.1 is [ABN24, Definition 23]. In particular, minimal codewords in additive codes are outer minimal. We decided to simply keep the word “minimal” to avoid to distinguish the case of $\mathbb{F}_{q^h}$ -linear codes.

Definition 3.3. We say that an additive code $\mathcal{C}$ is **minimal** if all its nonzero codewords are minimal.

Example 3.4. Consider $\mathbb{F}_4 = \mathbb{F}_2(\omega)$, with ω a root of $x^2 + x + 1 \in \mathbb{F}_2[x]$. Let $\mathcal{C}$ be the $[4, 2/2, 3]_2^2$ additive code defined with generator matrix

$$G = \begin{pmatrix} 1 & 1 & \omega & \omega \\ 1 & \omega & 0 & 1 \end{pmatrix}.$$

We have that $\mathcal{C} = \{(0, 0, 0, 0), (1, 1, \omega, \omega), (1, \omega, 0, 1), (0, \omega^2, \omega, \omega^2)\}$, and, hence, it is straightforward to see that the code $\mathcal{C}$ is minimal.

The following result is easy to verify, so we omit the proof.

Proposition 3.5. Let $\mathcal{C}, \mathcal{C}'$ be two $[n, k/h, d]_q^h$ equivalent additive codes. $\mathcal{C}$ is minimal if and only if $\mathcal{C}'$ is minimal.

We introduce the following object as generalization of outer strong blocking sets; see [ABN24, Definition 18].

Definition 3.6. An h - $[n, k, d]_q$ projective system $\{\pi_1, \dots, \pi_n\}$ is an **additive strong blocking set** in $\text{PG}(k-1, q)$ if $\bigcup_{i=1}^n \pi_i$ is a strong blocking set.

Remark 3.7. Additive strong blocking sets and outer strong blocking sets are related but distinct. Outer strong blocking sets arise from points in $\text{PG}(k/h-1, q^h)$ whose field reduction gives rise to subspaces of a Desarguesian spread, whereas additive strong blocking sets are defined more generally, requiring only that the union of their subspaces forms a strong blocking set in $\text{PG}(k-1, q)$. In particular, every outer strong blocking set gives an additive strong blocking set, but not every additive strong blocking set gives an outer strong blocking set.

The following result is straightforward, hence we omit its proof.

Proposition 3.8. Let χ and ψ be two equivalent h - $[n, k, d]_q$ projective systems. Then χ is an additive strong blocking set if and only if ψ is an additive strong blocking set.

In the rest of this section, we show that equivalence classes of minimal additive codes are in one-to-one correspondence with equivalence classes of additive strong blocking sets. First of all, observe that by Propositions 3.5 and 3.8, the properties of being minimal and additive strong blocking sets are invariant under equivalence. We introduce the following useful notation.

Notation 3.9. Let $\mathcal{C}$ an $[n, k/h, d]_q^h$ nondegenerate additive code. We define the $\mathbb{F}_q$ -linear k -dimensional code $\widehat{\mathcal{C}} \subseteq \mathbb{F}_q^{nq^h}$ as follows. Let G be a generator matrix of $\mathcal{C}$. Then, consider $\widetilde{G}$ as in Eq. (1). With a slight abuse of notation, we denote the columns of $\widetilde{G}$ by $(P_1^1, \dots, P_h^1 | \dots | P_1^n, \dots, P_h^n)$. We define by $\widetilde{\mathcal{C}}$ the linear code obtained as the $\mathbb{F}_q$ -span of the rows of $\widetilde{G}$. Now, we extend the matrix $\widetilde{G}$ to a matrix

$$\widehat{G} = \left(\widehat{G}_1 \mid \dots \mid \widehat{G}_n \right),$$

where each block $\widehat{G}_i$ consists of q^h columns, where the first h are $(P_1^i, \dots, P_h^i)$ and the others are all their $\mathbb{F}_q$ -linear combinations. Note that if those linear combinations are less than q^h , we add some zero columns. Finally, we define the code $\widehat{\mathcal{C}}$ as the $\mathbb{F}_q$ -span of the rows of $\widehat{G}$. Note that this procedure is well-defined, since all possible choices involved to construct $\widehat{G}$ lead to linear codes that are equivalent.

Lemma 3.10. Let $\mathcal{C}$ be a $[n, k/h, d]_q^h$ minimal additive code. Then $\widehat{\mathcal{C}} \subseteq \mathbb{F}_q^{nq^h}$ is a k -dimensional minimal $\mathbb{F}_q$ -linear code.

Proof. Suppose that $\mathcal{C}$ is minimal. Let $x := (x_1^1, \dots, x_{q^h}^1, \dots, x_1^n, \dots, x_{q^h}^n)$, $x' := (x_1^{1'}, \dots, x_{q^h}^{1'}, \dots, x_1^{n'}, \dots, x_{q^h}^{n'}) \in \widehat{\mathcal{C}} \setminus \{0\}$ such that $\text{supp}(x') \subseteq \text{supp}(x)$. Let us define their following corresponding codewords in $\widetilde{\mathcal{C}}$ and $\mathcal{C}$

$$\begin{aligned}\tilde{c} &:= (x_1^1, \dots, x_h^1, \dots, x_1^n, \dots, x_h^n) \in \widetilde{\mathcal{C}} \setminus \{0\}, \\ \tilde{c}' &:= (x_1^{1'}, \dots, x_h^{1'}, \dots, x_1^{n'}, \dots, x_h^{n'}) \in \widetilde{\mathcal{C}} \setminus \{0\}, \\ c &:= \left(\sum_{i=1}^h x_i^1 \alpha_i, \dots, \sum_{i=1}^h x_i^n \alpha_i \right) \in \mathcal{C} \setminus \{0\}, \\ c' &:= \left(\sum_{i=1}^h x_i^{1'} \alpha_i, \dots, \sum_{i=1}^h x_i^{n'} \alpha_i \right) \in \mathcal{C} \setminus \{0\}.\end{aligned}$$

We want that there exists a $\lambda \in \mathbb{F}_q$ such that $x' = \lambda x$, and this is equivalent to have $c' = \lambda c$, since $\mathcal{B} = \{\alpha_1, \dots, \alpha_h\}$ is a basis of $\mathbb{F}_{q^h}/\mathbb{F}_q$. First, since $\text{supp}(x') \subseteq \text{supp}(x)$, we have that $\text{supp}(\tilde{c}') \subseteq \text{supp}(\tilde{c})$. Therefore, $\text{supp}(c') \subseteq \text{supp}(c)$. Indeed, if $c_j = \sum_{i=1}^h x_i^j \alpha_i = 0$, then $\forall i \in \{1, \dots, h\}$, $x_i^j = 0$, since $\mathcal{B}$ is a basis of $\mathbb{F}_{q^h}/\mathbb{F}_q$. Thus, $\forall i \in \{1, \dots, h\}$, $x_i^{j'} = 0$, precisely because $\text{supp}(\tilde{c}') \subseteq \text{supp}(\tilde{c})$, and then $c'_j = \sum_{i=1}^h x_i^{j'} \alpha_i = 0$. Moreover, let $j \in \text{supp}(c')$ and let us show that $\frac{c'_j}{c_j} \in \mathbb{F}_q$, that is equivalent to show that there exists a $\lambda_j \in \mathbb{F}_q^*$ such that $\forall i \in \{1, \dots, h\}$, $x_i^{j'} = \lambda_j x_i^j$. Suppose by contradiction that this is not the case. Up to taking a code equivalent to $\widehat{\mathcal{C}}$, we have $x_1^1, x_2^1 \neq 0$, $x_1^{1'} = \lambda x_1^1$, and $x_2^{1'} = \mu x_2^1$ with $\lambda, \mu \in \mathbb{F}_q$ such that $\lambda \neq \mu$. Then:

- $x_1^1 x_2^1 - x_2^1 x_1^1 = 0$ which is a coefficient of the codeword x in some coordinate ℓ .
- $x_1^1 x_2^{1'} - x_2^{1'} x_1^{1'} = (\mu - \lambda)(x_1^1 x_2^1) \neq 0$ which is a coefficient of the codeword x' in the same coordinate ℓ as before.

Therefore, $\text{supp}(x') \not\subseteq \text{supp}(x)$ and we have our contradiction. To conclude, since $\mathcal{C}$ is a minimal additive code, there exists a $\lambda \in \mathbb{F}_q$ such that $c' = \lambda c$ and then $x' = \lambda x$. Hence, the linear code $\widehat{\mathcal{C}}$ is minimal. $\square$

The following is the main result of this section.

Theorem 3.11. Equivalence classes of $[n, k/h, d]_q^h$ nondegenerate minimal additive codes are in one-to-one correspondence with equivalence classes of h - $[n, k/h, d]_q$ projective systems that are additive strong blocking sets.

Proof. The statement of the theorem is well-defined because of Propositions 3.5 and 3.8. Let $\mathcal{C}$ be an $[n, k/h, d]_q^k$ be an additive code and let $\chi = \{\pi_1, \dots, \pi_n\}$ be its associated h -projective system. Let $G = (g_1 \mid \dots \mid g_n)$ be a generator matrix of $\mathcal{C}$.

If $\mathcal{C}$ is minimal, then, by Lemma 3.10, $\widehat{\mathcal{C}}$ is a minimal linear code, hence an associated projective system $\mathcal{P}(\widehat{\mathcal{C}})$ is a strong blocking set (see e.g. [ABN22, Theorem 3.4]). Since as sets, $\mathcal{P}(\widehat{\mathcal{C}}) = \cup_{i=1}^n \pi_i$, we have that χ is an additive strong blocking set.

Conversely, suppose that χ is an additive strong blocking set. Then, for all hyperplanes H, H' of $\text{PG}(k-1, q)$ we have

$$\cup \pi_i \cap H \subseteq \cup \pi_i \cap H' \Leftrightarrow H = H' \quad (2)$$

Let $c = uG, c' = u'G$ be two nonzero codewords of $\mathcal{C}$ such that $\text{supp}(c') \subseteq \text{supp}(c)$ and $\forall i \in \text{supp}(c'), \frac{c'_i}{c_i} \in \mathbb{F}_q^*$. We have to show the existence of a $\lambda \in \mathbb{F}_q^*$ such that $c' = \lambda c$, that is equivalent to show that the two hyperplanes $\langle u \rangle^\perp$ and $\langle u' \rangle^\perp$ are equal. Since $\text{supp}(c') \subseteq \text{supp}(c)$, we have the inclusion $\{i \in \{1, \dots, n\} : \pi_i \subseteq \langle u' \rangle^\perp\} \subseteq \{i \in \{1, \dots, n\} : \pi_i \subseteq \langle u \rangle^\perp\}$. Moreover, $\forall i \in \text{supp}(c')$, we have that $\frac{c'_i}{c_i} \in \mathbb{F}_q^*$, hence for all i such that $\pi_i \not\subseteq \langle u' \rangle^\perp$, we have the existence of a $\lambda \in \mathbb{F}_q^*$ such that $\pi_i \subseteq \langle u' - \lambda u \rangle^\perp$. Now, let $i \in \{1, \dots, n\}$. If $\pi_i \cap \langle u \rangle^\perp = \pi_i$, then $\pi_i \subseteq \langle u \rangle^\perp$, and, therefore, $\pi_i \subseteq \langle u' \rangle^\perp$. Thus, $\pi_i \cap \langle u \rangle^\perp = \pi_i \cap \langle u' \rangle^\perp = \pi_i$. Otherwise, if $\pi_i \cap \langle u \rangle^\perp \neq \pi_i$, let us denote $\pi_i \cap \langle u \rangle^\perp = \{g_1, \dots, g_p\}$. Since there exists a $\lambda \in \mathbb{F}_q^*$ such that $\pi_i \subseteq \langle u' - \lambda u \rangle^\perp$, we have that $g_1, \dots, g_p \in \langle u' - \lambda u \rangle^\perp$. But $g_1, \dots, g_p \in \langle u \rangle^\perp$, hence $g_1, \dots, g_p \in \langle u' \rangle^\perp$ and then $g_1, \dots, g_p \in \pi_i \cap \langle u' \rangle^\perp$. Therefore, in any case, we have that $\pi_i \cap \langle u \rangle^\perp = \pi_i \cap \langle u' \rangle^\perp$. It is satisfied for all i , thus we have that $\cup \pi_i \cap \langle u \rangle^\perp = \cup \pi_i \cap \langle u' \rangle^\perp$ and then, by Eq. (2), $\langle u \rangle^\perp = \langle u' \rangle^\perp$. Hence, $\mathcal{C}$ is a minimal additive code. $\square$

Example 3.12. Consider $\mathbb{F}_4 = \mathbb{F}_2(\omega)$, with ω a root of $x^2 + x + 1 \in \mathbb{F}_2[x]$. Let $\mathcal{C}$ be the $[3, 3/2, 2]_2^2$ code defined by the matrix :

$$G = \begin{pmatrix} 1 & 1 & 1 \\ \omega & \omega^2 & \omega \\ 0 & \omega & \omega \end{pmatrix}.$$

Now, we choose $\{1, \omega\}$ as a basis of $\mathbb{F}_4$ over $\mathbb{F}_2$. The extended generator matrix of $\mathcal{C}$ is

$$\tilde{G} = \left(\begin{array}{cc|cc|cc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{array} \right).$$

Thus, an 2 - $[3, 3, 2]_2$ projective system associated to $\mathcal{C}$ is $\chi := (\pi_1, \pi_2, \pi_3)$, where

- $\pi_1 = \{[1 : 0 : 0], [0 : 1 : 0], [1 : 1 : 0]\},$

- $\pi_2 = \{[1 : 1 : 0], [0 : 1 : 1], [1 : 0 : 1]\},$
- $\pi_3 = \{[1 : 0 : 0], [0 : 1 : 1], [1 : 1 : 1]\}.$

Hence, $\cup_{i=1}^3 \pi_i$ is the union of 3 distinct lines of $\text{PG}(2, 2)$ which form a strong blocking set (see e.g. [ABN22, Theorem 5.3]), hence χ is an additive strong blocking set in $\text{PG}(2, 2)$. Therefore, the code $\mathcal{C}$ is minimal.

4 Existence of minimal additive codes

In this section, we will use Definition 3.1 to derive an existence condition for minimal additive codes. This result generalizes [ABN24, Theorem 38] for outer minimal codes.

Theorem 4.1. If

$$\binom{nh-2}{k-2}_q \cdot \sum_{i=1}^n \binom{n}{i} (q^h - 1)^i (q^i - q) < \binom{nh}{k}_q,$$

then there exists an $[n, k/h]_q^h$ minimal additive code.

Proof. An $[n, k/h]_q^h$ additive code $\mathcal{C}$ is not minimal if it contains a codeword which is not minimal. A nonzero codeword $c \in \mathcal{C} \subseteq \mathbb{F}_{q^h}^n$ is not minimal if there exists $c' \in \mathcal{C}$ such that it holds

$$\begin{cases} c' \neq \lambda c, \forall \lambda \in \mathbb{F}_q \wedge \\ \text{supp}(c') \subseteq \text{supp}(c) \wedge \\ \forall i \in \text{supp}(c), \exists \lambda_i \in \mathbb{F}_q \text{ s.t. } c'_i = \lambda_i c_i. \end{cases} \quad (3)$$

Let

$$B = \left\{ (c, c') \in \mathbb{F}_{q^h}^n \times \mathbb{F}_{q^h}^n : c \neq 0 \text{ and (3) holds} \right\}.$$

The cardinality of B is given by

$$|B| = \sum_{i=1}^n \binom{n}{i} (q^h - 1)^i (q^i - q).$$

An $[n, k/h]_q^h$ additive code is not minimal if it contains the $\mathbb{F}_q$ -subspace generated one of the pairs in B . Since the 2-dimensional $\mathbb{F}_q$ -subspace generated by each pair in B is contained in exactly $\binom{nh-2}{k-2}_q$ additive codes of dimension k in $\mathbb{F}_{q^h}^n$ and the total number of $[n, k/h]_q^h$ codes is $\binom{nh}{k}_q$, we have that if $\binom{nh-2}{k-2}_q \cdot |B| < \binom{nh}{k}_q$, there exists an $[n, k/h]_q^h$ additive code $\mathcal{C}$ which does not contain any 2-dimensional $\mathbb{F}_q$ subspace generated by a pair in B . Therefore, such $\mathcal{C}$ must be minimal. $\square$

Corollary 4.2. $[n, k/h]_q^h$ minimal additive codes exist whenever

$$n \geq \left\lceil \frac{2k}{h \log_{q^h} \left(\frac{q^{2h}}{q^{h+1}-q+1} \right)} \right\rceil.$$

Proof. First of all observe that

$$|B| < (1 + (q^h - 1)q)^n$$

and

$$\frac{\binom{nh}{k}_q}{\binom{nh-2}{k-2}_q} = \frac{(q^{nh} - 1)(q^{nh-1} - 1)}{(q^k - 1)(q^{k-1} - 1)} \geq q^{2nh-2k}.$$

In order to ensure the existence of minimal additive codes, from Theorem 4.1, it is enough to have

$$(1 + (q^h - 1)q)^n < q^{2nh-2k}.$$

Taking logarithms base q^h yields

$$n \log_{q^h} (q^{h+1} - q + 1) < 2 \left(n - \frac{k}{h} \right).$$

Rearranging, we obtain

$$n(2 - \log_{q^h} (q^{h+1} - q + 1)) > \frac{2k}{h}.$$

Equivalently,

$$n \geq \left\lceil \frac{2k}{h \log_{q^h} \left(\frac{q^{2h}}{q^{h+1}-q+1} \right)} \right\rceil.$$

□

Note that the lower bound in [ABN24, Theorem 38] that ensures the existence of outer minimal linear codes is larger than the one provided in Corollary 4.2, for every $h \geq 2$. Hence, in general, minimal additive codes exist also when outer minimal linear codes do not necessarily exist.

5 Conclusion

In this paper, we introduced minimal additive codes and their geometric counterpart, additive strong blocking sets, as a natural generalization of outer minimal codes and outer strong blocking sets. We established a one-to-one correspondence between equivalence classes of minimal additive codes and additive strong blocking sets, and provided a first sufficient condition for their existence. This geometric perspective lays the groundwork for further study of minimal additive codes, including construction, parameter restrictions, and potential applications in secret sharing and quantum coding, which we plan to investigate in future research.

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